Power Budget - Logan Maelstrom (Individual Subsystem)

Team Number: 104
Project Name: **Drowsy Driver Alarm**
Name: **Logan Maelstrom**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity N	DM164150	1.8V - 5.5V	1	13.8	13.8	mA
	BMI270 6-Axis IMU	BMI270	1.7V - 3.6V	1	0.97	0.97	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity N	DM164150	1.8V - 5.5V	1	13.8	13.8	mA
					Subtotal	13.8	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	17.25	mA
Regulator	+5V Regulator	L7805ACV-DG	7V - 35V	1	1000	1000	mA
					Total Remaining Current Available on +5V Rail	982.75	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	BMI270 6-Axis IMU	BMI270	1.7V - 3.6V	1	0.97	0.97	mA
					Subtotal	0.97	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	1.2125	mA
Regulator	+3.3V LDO Regulator	RT9080-33GJ5	1.2V - 5.5V	1	600	600	mA
					Total Remaining Current Available on 3.3V Rail	598.7875	mA

External Power Source	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source Selection	9V External Wall Supply	Model 0930	100–240 VAC	9 VDC	3000	3000	mA
Power Rails Connected to External Power Source	+5V regulator	L7805ACV-DG	7V - 35V	1	1000	1000	mA
Total Remaining Current Available on External Power Source 1						2000	mA